

| Calculations                                                         |          |          |           |           |
|----------------------------------------------------------------------|----------|----------|-----------|-----------|
|                                                                      | Sheep    | Pig      | Cow       | Goat      |
| Meat Diet Portion                                                    | 1.07%    | 10.89%   | 14.20%    | 3.55%     |
| Meat MCal Target                                                     | 972.93   | 9,932.61 | 12,957.11 | 3,238.60  |
| Potential Dairy Value attached to each juvenile                      | 132.54   | 0.00     | 1015.20   | 126.90    |
| Value per juvenile                                                   | 48.00    | 160.00   | 640.00    | 48.00     |
| Adjusted Dairy Value attached to juvenile based on Dairy Utilization | 0.00     | 0.00     | 0.00      | 0.00      |
| Culled Mothers Value                                                 | 14.77    | 4.41     | 105.26    | 14.77     |
| Culled Adult Males                                                   | 0.74     | 0.22     | 5.26      | 0.74      |
| Final Value per juvenile                                             | 63.51    | 164.63   | 750.53    | 63.51     |
| Value MCals                                                          | 63.51    | 164.63   | 750.53    | 63.51     |
| Offspring required Year <sup>-1</sup>                                | 15.32    | 60.33    | 17.26     | 51.00     |
| MCals from meat                                                      | 972.93   | 9932.61  | 12957.11  | 3238.60   |
| Max Utilized Mcals allowed from Dairy                                | 0.00     | 0.00     | 0.00      | 0.00      |
| MCals actually Utilized from Dairy                                   | 0.00     | 0.00     | 0.00      | 0.00      |
| Percent of diet from their meat                                      | 1.07%    | 10.89%   | 14.20%    | 3.55%     |
| Percent of Diet from their Dairy                                     | 0.00%    | 0.00%    | 0.00%     | 0.00%     |
| Kg Fleece                                                            | 117.30   | 0.00     | 0.00      | 0.00      |
| 1 Year+ (age) skins                                                  | 5.99     | 2.59     | 3.50      | 19.25     |
| Yearly Offspring skins                                               | 28.47    | 71.68    | 25.80     | 93.38     |
| Birthing Events per year                                             | 1.00     | 1.86     | 1.00      | 1.00      |
| Offspring Mother <sup>-1</sup> Year <sup>-1</sup>                    | 0.56     | 5.48     | 0.68      | 0.56      |
| Mothers Step 1                                                       | 27.36    | 11.02    | 25.39     | 91.06     |
| Males Step 1                                                         | 7.66     | 30.17    | 8.63      | 25.50     |
| Females Step 1                                                       | 7.66     | 30.17    | 8.63      | 25.50     |
| Replacement Mothers per year                                         | 5.77     | 2.50     | 3.36      | 18.51     |
| Breeding Males Required                                              | 1.44     | 0.58     | 1.33      | 4.78      |
| Additional Mothers                                                   | 23.48    | 2.07     | 12.55     | 75.68     |
| Males Step 2                                                         | 6.58     | 5.67     | 4.27      | 21.19     |
| Females 2                                                            | 6.58     | 5.67     | 4.27      | 21.19     |
| Total Mothers                                                        | 33.13    | 13.52    | 28.75     | 109.58    |
| Total Males                                                          | 14.24    | 35.84    | 12.90     | 46.69     |
| Total Females                                                        | 1.08     | 24.50    | 4.36      | 4.31      |
| Total Offspring                                                      | 28.47    | 71.68    | 25.80     | 93.38     |
| Total Animals                                                        | 49.88    | 74.44    | 47.35     | 165.35    |
| People per person                                                    | 1.00     |          |           |           |
| Total Adults per Hectare                                             | 1.77     | 0.27     | 0.43      | 1.76      |
| Total Offspring per Hectare                                          | 0.87     | 1.18     | 0.26      | 0.86      |
|                                                                      | 0.75%    |          | 9.05%     | 2.46%     |
| Total Adults                                                         | 58       | 16       | 43        | 190       |
| Total Offspring                                                      | 28       | 72       | 26        | 93        |
| Days for Maintenance                                                 | 5.00     | 220.00   | 0.00      | 5.00      |
| Dry Mothers                                                          | 4.66     | 0.00     | 2.95      | 16.20     |
| Other                                                                | 455.54   | 0.00     | 0.00      | 1,494.00  |
| MCals Dry Mothers                                                    | 5,439.49 | 0.00     | 6,246.19  | 18,922.36 |
| total Maintenance MCals                                              | 5,895.03 | 0.00     | 6,246.19  | 20,416.36 |

| MCals Required |          |           |           |           |
|----------------|----------|-----------|-----------|-----------|
|                | Sheep    | Pig       | Cow       | Goat      |
| Gestating      | 16,955   | 32,724    | 43,187    | 55,605    |
| Lactating      | 32,346   | 11,633    | 46,438    | 106,084   |
| Maintenance    | 5,895    | 0         | 6,246     | 20,416    |
| Adult Males    | 1,678    | 0         | 2,822     | 5,584     |
| Offspring      | 7,815    | 72,036    | 94,424    | 25,631    |
| TOTAL          | 64,689.0 | 116,393.2 | 193,116.7 | 213,320.3 |

| Animal Diet Breakdown           |        |         |         |         |
|---------------------------------|--------|---------|---------|---------|
|                                 | Sheep  | Pig     | Cow     | Goat    |
| Crop                            | Barley | Barley  | Barley  | Barley  |
| Grams <sup>-1</sup> Straw/Chaff | 1000.0 | 0.0     | 4956.0  | 1000.0  |
| MCal kg <sup>-1</sup>           |        |         |         |         |
| Grams <sup>-1</sup> Grain       | 750.0  | 750.0   | 1500.0  | 750.0   |
| MCal kg <sup>-1</sup>           |        |         |         |         |
| Days                            | 90.0   | 90.0    | 90.0    | 90.0    |
| Number of Animals               | 28.5   | 71.7    | 25.8    | 93.4    |
| MCals from Straw/Chaff          | 3638.6 | 0.0     | 16340.4 | 11933.3 |
| MCals from Grain                | 5765.5 | 14514.7 | 10448.5 | 18908.5 |
| MCals from other feed           | 9404.1 | 14514.7 | 26788.9 | 30841.8 |

| Diet Calculations                           |            |         |
|---------------------------------------------|------------|---------|
| Plants Mcal                                 | 63,875.000 |         |
| Meat Mcal                                   | 27,375     |         |
| Dairy                                       | 0          |         |
|                                             | Perfect    |         |
| Plants %                                    | 70.00%     |         |
| Meat %                                      | 30.00%     |         |
| Dairy %                                     | 0.00%      |         |
| check                                       | Perfect    |         |
| Percent                                     |            |         |
| Domestic Meat                               | 29.70%     |         |
| Dairy                                       | 0.00%      |         |
| Wild Meat                                   | 0.30%      |         |
| Crops                                       | 63.00%     |         |
| Wild Plants                                 | 7.00%      |         |
| check                                       | Perfect    |         |
| new dairy                                   | 0.00       |         |
| potential dairy < limit                     | TRUE       |         |
| new limit                                   | 70.0%      |         |
| the limit greater than that left to satisfy | TRUE       |         |
| limit                                       | 1          |         |
| limit dairy checked                         | FALSE      |         |
| Mcalories                                   |            | Surplus |
| Domestic Meat                               | 27,101.250 |         |
| Dairy                                       | 0.000      |         |
| Wild Meat                                   | 273.750    |         |
| Crops                                       | 57,487.500 |         |
| Wild Plants                                 | 6,387.500  |         |
| check                                       | Perfect    |         |
| SURPLUS DAIRY                               | 0.000      |         |
| This could account for additional % of diet | 0.00%      |         |

Herd Modelling

|                                                               | Sheep | Pig    | Cow    | Goat   |  |  |  |  |
|---------------------------------------------------------------|-------|--------|--------|--------|--|--|--|--|
| Mothers Req'd<br>Males for Slaughter<br>Females for Slaughter | 33    | 14     | 29     | 110    |  |  |  |  |
|                                                               | 14    | 36     | 13     | 47     |  |  |  |  |
|                                                               | 1     | 24     | 4      | 4      |  |  |  |  |
| Offspring available for Consumption                           | 28    | 72     | 26     | 93     |  |  |  |  |
| Gestation Req'd                                               | 9123  | 27546  | 28900  | 30368  |  |  |  |  |
| Lactation Req'd                                               | 17405 | 9792   | 31075  | 57936  |  |  |  |  |
| Maintenance Req'd                                             | 21045 | 0      | 8894   | 69238  |  |  |  |  |
| Adult Males                                                   | 1678  | 0      | 2822   | 5584   |  |  |  |  |
| Offspring Required                                            | 9353  | 78487  | 112999 | 30674  |  |  |  |  |
| Energy Required                                               | 58604 | 115825 | 184690 | 193799 |  |  |  |  |
| Ewe Fleece Kg Yr-1                                            | 75    |        |        |        |  |  |  |  |
| Ram Fleece Kg year-1                                          | 3     |        |        |        |  |  |  |  |
| Juvenile Fleece Kg Yr-1                                       | 40    |        |        |        |  |  |  |  |
| Kg Fleece Year-1                                              | 117   | 0      | 0      | 0      |  |  |  |  |